

Random Transverse Field Ising Model

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We investigate the behavior of the random transverse field Ising model (RTFIM) by performing Monte-Carlo simulation. We choose the dimension of the system to be one ($d = 1$) for which the quantum fluctuations are enhanced. Because there is no frustration in our system, we are able to use highly efficient cluster algorithms which substantially reduces the critical slowing down.

The Hamiltonian of the random transverse field Ising model is given by

$$H = - \sum_{i=1}^L J_i \sigma_{\alpha,i}^z \sigma_{\alpha,i+1}^z - \sum_{i=1}^L h_i \sigma_{\alpha,i}^x \quad (1)$$

where L is the length of the lattice, Latin sub-indices denote the lattice sites, and σ 's are the Pauli operators. The J_i and h_i are the random nearest-neighbor coupling constants and the random transverse fields, respectively. The random coupling constants and transverse fields are taken from a distribution restricted to only positive values.

To study the Hamiltonian in Eq. (1), we propose an effective classical action in $d + 1$ dimensions, where the extra imaginary time dimension is of size $\beta \equiv 1/T$ and is divided up into $L_\tau \equiv \beta/\Delta\tau$ intervals each of width $\Delta\tau$ in the limit $\Delta\tau \rightarrow 0$. We introduce disorder only in the horizontal direction. The vertical bonds are pure and the horizontal bonds are the same within each column. This emulates a quenched disordered quantum system whose disorder is known to be perfectly correlated in the imaginary time direction. This is the McCoy-Wu random Ising model [1–4] which is shown to be equivalent to the random transverse field quantum spin- $\frac{1}{2}$ Ising model in the large imaginary time limit. The partition function is then

$$Z = \lim_{\Delta\tau \rightarrow 0} \text{Tr} e^{-S} \quad (2)$$

with the effective classical action given by

$$S_{MW} = - \sum_{\tau,i} J_i S_i(\tau) S_{i+1}(\tau) - \sum_{\tau,i} J S_i(\tau) S_i(\tau + 1) \quad (3)$$

where the $S_i(\tau) = \pm 1$ are classical Ising spins, the indices α and β denote the color, the index i runs over the sites of the original d -dimensional lattice, and $\tau = 1, 2, \dots, L_\tau$ denotes a time slice. Note that we have the same random interaction in each slice. For computational convenience, we set $\Delta\tau = 1$ and equivalently take the limit $L_\tau \rightarrow \infty$ implying $T \rightarrow 0$. The coupling J which sets the relative strength of the transverse field in Eq. (1), is like an inverse “temperature” for the effective classical model in Eq. (3). The spin-spin coupling,

J_i , is independent of τ , because it is a quenched random variable. Because the effect of disorder in a $d = 1$ dimensional quantum system is already more enhanced than any higher dimensional system, even a relatively strong disorder distribution should suffice to capture their effect on a quantum system. We independently take the couplings J_i from the following rectangular distributions

$$\pi(J_i) = \begin{cases} 1, & \text{if } J - \frac{\Delta J}{2} < J_i < J + \frac{\Delta J}{2} \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

Thus, the random couplings J_i have the mean value J and the width ΔJ . We use J as tuning parameter in our simulation.

We have studied the model (3) in $d = 1$ dimension by Monte Carlo simulations on a square lattice of size $L \times L_\tau$ using a periodic boundary conditions. We have estimated the equilibration “time” using the logarithmic binning method i.e. we have compared the average values of each observable over 2^n Monte Carlo steps and made sure that the last three averages were within each others error bars. Each observable has been obtained by averaging over 10,000 disordered configurations and for each disordered configuration, 10,000 thermal averages has been conducted. The error bars have been calculated using the Jackknife procedure [5–7].

The model in Eq. (1) is known to exhibit unusual behavior associated with infinite randomness fixed point and is described by strongly disordered critical points [8, 9]. This system undergoes a phase transition at zero temperature from a phase with infinite susceptibility and continuously varying exponents to a ferromagnetic phase via a quantum critical point characterized by scaling with energy E and length scales L related by $\ln E \sim L^\psi$.

The critical point of the classical McCoy-Wu model is well known [1–3] and it is given by

$$2J + \int_0^\infty dJ' P(J') \ln(\tanh(J')) = 0. \quad (5)$$

For the rectangular distribution given in Eq. 4 with $\Delta J = 0.2$, the critical point, according to Eq. (5), is $J_c = 0.450341$.

Now we try to calculate the critical point using our algorithm. We calculate the magnetic Binder cumulant [10, 11]

$$V_m = 1 - \frac{[\langle m^4 \rangle]}{3[\langle m^2 \rangle]^2} \quad (6)$$

where

$$m = \frac{1}{LL_\tau} \sum_{\tau,i} S_i(\tau). \quad (7)$$

The square and angular brackets, $[\dots]$ and $\langle \dots \rangle$, denote the disordered and temporal averages, respectively. In the disordered phase, the probability distribution of the magnetization, $P(m)$, becomes a Gaussian and $V_m \rightarrow 0$ as $L \rightarrow \infty$. In the ordered phase, $P(m)$ becomes a δ -function and $V_m \rightarrow 2/3$ as $L \rightarrow \infty$. In the paramagnetic phase, for small L_τ , the system is effectively classical and at a finite temperature, therefore $V_m \rightarrow 0$. For $L_\tau \rightarrow \infty$, the system is quasi one dimensional in the imaginary time direction, therefore $V_m \rightarrow 0$ also. There exists an intermediate point where the V_m acquires a maximum value $V_m^{\max}$. This maximum value decreases as L increases if the system is in the paramagnetic phase, whereas it increases as L increases if the system is in the ferromagnetic phase. So, there is an intermediate point at which the $V_m^{\max}$ is a constant for all L which is the quantum critical point. For our model with $\Delta_J = 0.20$ we estimate the quantum critical point $J_c = 0.445 \pm 0.005$.

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